

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Following are the functional requirements of the proposed Incident Lifecycle Automation solution.

Date	08 September 2026
Team ID	
Project Name	Incident Lifecycle Automation in ServiceNow
Maximum Marks	4 Marks

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Service & Service Offering	Create/select the required Service and Service Offering for incident classification and support context.
FR-2	Incident Record Creation	Create an Incident Record with the required incident information and standard lifecycle fields.
FR-3	Incident Classification	Classify incidents using Category, Subcategory, Urgency, Service and Configuration Item (CI).
FR-4	Knowledge Management & Agent Assist	Search and use Knowledge Management and Agent Assist capabilities to support troubleshooting and resolution.
FR-5	Reassignment & Escalation	Reassign or escalate unresolved incidents to the appropriate support team / Level 2 support.
FR-6	Level 2 Investigation	Allow Level 2 technicians to investigate, track work and maintain investigation context on escalated incidents.
FR-7	Emergency Change	Create/link an Emergency Change Request when incident resolution requires an urgent change.
FR-8	Child Incident	Create and relate Child Incidents for related or recurring issues while retaining parent incident context.
FR-9	Resolution & Closure	Capture Cause, Resolution, Resolution Details and closure information before completing the incident.
FR-10	Knowledge Article Creation	Create a Knowledge Article from useful troubleshooting or resolution information for future reuse.
FR-11	SLA & Related Records	Provide visibility of SLA and related records such as changes, child incidents and knowledge.
FR-12	Final Validation & Testing	Validate the incident, service, SLA, change, child incident, knowledge and related records and perform testing/deployment validation.

Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a clear, consistent Incident Management workflow so Service Desk and Level 2 users can complete tasks with minimal confusion.
NFR-2	Security	Use ServiceNow roles, ACLs and least-privilege access so users can access only the incident and related functions appropriate to their role.

NFR-3	Reliability	Maintain consistent incident, related-record and resolution data throughout reassignment, escalation, change and closure activities.
NFR-4	Performance	Incident forms, searches, related records and knowledge access should respond efficiently under normal operational load.
NFR-5	Availability	Use the ServiceNow platform capabilities to support continuous access to incident management and related records, subject to platform availability.
NFR-6	Scalability	Support growth in incident volume, users, services, configuration items, knowledge articles and related records without redesigning the core lifecycle.

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The solution is implemented as a ServiceNow-centered architecture. Users interact through the ServiceNow interface; Incident Management coordinates classification and lifecycle activities; Service/Service Offering, CI, Knowledge, Change Management, parent/child incidents, SLA and resolution records provide the supporting capabilities.

Incident Lifecycle Automation in ServiceNow - Technical Architecture

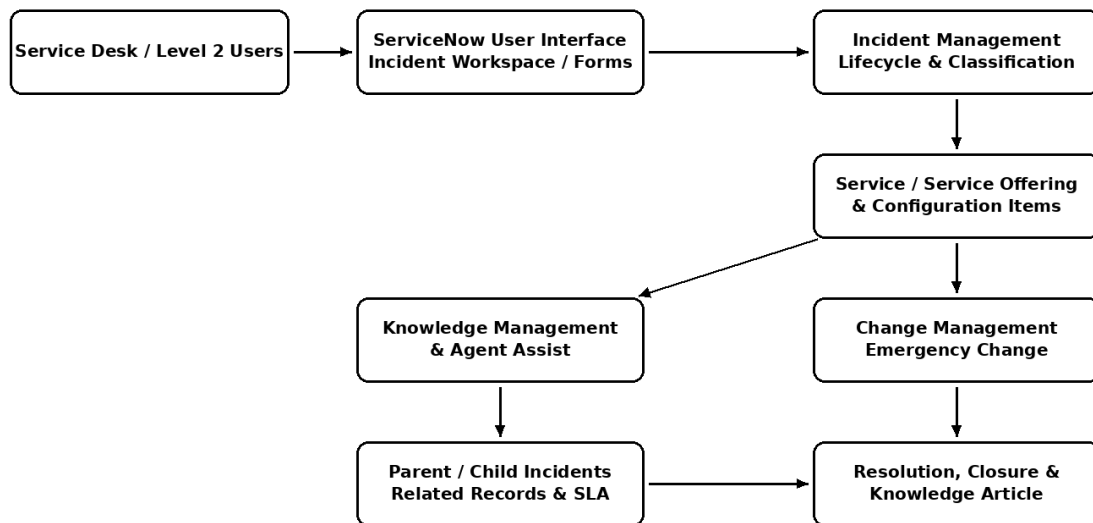


Table-1 : Components & Technologies:

Date	08 September 2026
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S.No	Component	Description	Technology
1	User Interface	Service Desk Agent and Level 2 Technician interact with incident forms/workspaces and related-record views.	ServiceNow UI / Incident Workspace
2	Application Logic-1	Incident creation, classification, assignment, reassignment, escalation and lifecycle processing.	ServiceNow Incident Management
3	Application Logic-2	Service, Service Offering and CI provide operational context for the incident.	ServiceNow Service / Service Offering / CMDB

4	Application Logic-3	Knowledge search and Agent Assist support troubleshooting and resolution.	ServiceNow Knowledge Management / Agent Assist
5	Database	Stores incidents, users, services, CIs, work notes, resolution information and related records.	ServiceNow platform database
6	Cloud Database	Platform-managed data persistence for ServiceNow application data.	ServiceNow cloud platform
7	File Storage	Attachments such as supporting evidence can be associated with records.	ServiceNow attachment storage
8	External API-1	No project-specific external API is required for the core lifecycle described.	Not applicable
9	External API-2	No project-specific external API is required for the core lifecycle described.	Not applicable
10	Change / Related Records	Emergency Change Requests and Child Incidents are created/linked to maintain traceability.	ServiceNow Change Management / Incident relationships
11	SLA Management	SLA information is associated with the incident for operational visibility and validation.	ServiceNow SLA / Task SLA capabilities

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The project uses ServiceNow platform capabilities rather than a separate open-source application framework.	ServiceNow platform
2	Security Implementations	Role-based access and ServiceNow access controls should restrict incident and related-record operations according to user responsibility.	ServiceNow roles / ACLs
3	Scalable Architecture	Platform services separate user interaction, application capabilities and managed data, supporting growth in incidents and related records.	ServiceNow cloud architecture
4	Availability	The solution relies on the ServiceNow platform for hosted service availability and centralized access.	ServiceNow cloud platform
5	Performance	Efficient forms, indexed record searches and controlled related-record/knowledge queries should be used for operational responsiveness.	ServiceNow platform capabilities

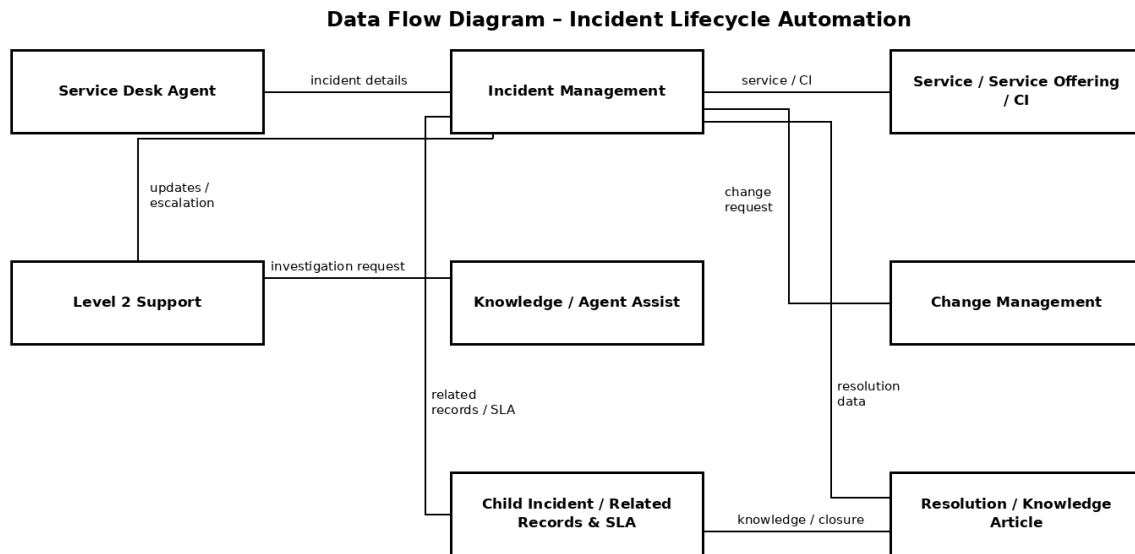
References: The original template references C4 Model, IBM Cloud Architecture, AWS Architecture and architecture-diagram guidance. These references are retained as general architecture guidance; no external technology beyond the ServiceNow platform is assumed for the project.

Project Design Phase-II

Data Flow Diagram & User Stories

Data Flow Diagrams:

The DFD below shows how incident information enters the Incident Management process, how it is enriched with service/CI and knowledge information, and how it flows through Level 2 support, change management, related incidents, SLA tracking, resolution and knowledge creation.



User Stories

Use the below template to list all the user stories for the product.

Date	08 September 2026
Team ID	
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Maximum Marks	4 Marks

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Service Desk Agent	Service & Service Offering	USN-1	As a Service Desk Agent, I can select the appropriate Service and Service Offering when working on an incident.	The incident contains the required service context.	High	Sprint-1
Service Desk Agent	Incident Creation	USN-2	As a Service Desk Agent, I can create an Incident Record with complete	A new incident is created with the required fields.	High	Sprint-1

			incident information.			
Service Desk Agent	Incident Classification	USN-3	As a Service Desk Agent, I can classify an incident using Category and Subcategory.	The selected classification is saved correctly.	High	Sprint-1
Service Desk Agent	Incident Classification	USN-4	As a Service Desk Agent, I can record Urgency, Service and Configuration Item (CI).	Urgency, service and CI are available and saved on the incident.	High	Sprint-1
Service Desk Agent	Knowledge Management	USN-5	As a Service Desk Agent, I can use Knowledge Management and Agent Assist to find troubleshooting guidance.	Relevant knowledge can be searched and used during investigation.	High	Sprint-1
Service Desk Agent	Reassignment / Escalation	USN-6	As a Service Desk Agent, I can reassign or escalate an unresolved incident to Level 2 support.	The incident is routed to the appropriate Level 2 support team with context.	High	Sprint-1
Level 2 Technician	Level 2 Investigation	USN-7	As a Level 2 Technician, I can investigate an escalated incident and track investigation work.	Investigation updates are recorded and visible on the incident.	High	Sprint-2
Level 2 Technician	Emergency Change	USN-8	As a Level 2 Technician, I can create or link an Emergency Change Request when an urgent change is required.	The change is related to the incident for traceability.	Medium	Sprint-2
Service Desk Agent	Child Incident	USN-9	As a Service Desk Agent, I can create Child Incidents for related or recurring issues.	Child incidents are linked to the parent incident and retain the relationship.	Medium	Sprint-2
Level 2 Technician	Resolution & Closure	USN-10	As a Level 2 Technician, I can record Cause, Resolution and Resolution Details and close the incident.	Required resolution and closure information is captured before closure.	High	Sprint-2
Service Owner / Support Stakeholder	Knowledge Article	USN-11	As a support stakeholder, I can create a Knowledge Article from a useful incident resolution.	A reusable article can be created from the resolved knowledge.	Medium	Sprint-2
Project Team	Validation & Testing	USN-12	As a project team member, I can validate the incident, service, SLA, change, child incident, knowledge and related records.	All required related records and lifecycle steps are validated successfully.	High	Sprint-2

Requirement Analysis – Customer Journey Map

Incident Lifecycle Automation in ServiceNow

Date 08 September 2026	Team ID	Primary Customer Service Desk Agent	Supporting Customer Level 2 Technician
Scenario Creating, investigating, escalating, resolving and validating an incident	Project Incident Lifecycle Automation in ServiceNow	Journey Type ServiceNow Incident Management lifecycle	Reference Format Customer Journey Map example from uploaded Requirement Analysis folder

Customer Journey Map – Incident Lifecycle Automation

Primary customer: Service Desk Agent | Supporting customer: Level 2 Technician

CUSTOMER JOURNEY	ENTICE <i>Discover / understand the support process</i>	ENTER <i>Create and classify the incident</i>	ENGAGE <i>Investigate, escalate and connect related work</i>	EXIT <i>Resolve, document and close</i>	REUSE <i>Reuse knowledge and improve future support</i>
STEPS	Identify a user issue and support need.	Create an Incident Record and select Service / Service Offering; classify Category, Subcategory, Urgency and CI.	Use Knowledge / Agent Assist; reassign or escalate; Level 2 investigates; create Emergency Change or Child Incident if required.	Record Cause, Resolution, Resolution Details and closure information.	Create a Knowledge Article from the resolution information; validate the incident, SLA, change, child incident and knowledge records before deployment.
INTERACTIONS	Service Desk Agent receives the issue and determines that it needs structured Incident Management.	Agent enters incident details and classification information in ServiceNow.	Agent searches knowledge and updates the incident; Level 2 reviews history, work notes, knowledge and related records.	Level 2 / support team completes the fix and records resolution details.	Support stakeholders review the knowledge article; the team performs final validation and testing before closure.
TOUCHPOINTS	ServiceNow Incident Management process; Service Desk communication.	Incident form; Service; Service Offering; Category; Subcategory; Urgency; CI.	Knowledge Management; Agent Assist; assignment group; Level 2 queue; Change Request; Parent/Child Incident links; SLA.	Resolution fields; closure information; incident activity/history.	Knowledge Article record; related-record views; validation and testing results.
GOALS & MOTIVATIONS	Help me understand how to start and capture the issue correctly.	Help me create a complete, correctly classified and traceable incident.	Help me find a solution quickly and make sure the right technical team has full context.	Help me resolve the issue and leave clear documentation for stakeholders.	Help me preserve reusable knowledge and confirm the full lifecycle closed correctly.
EMOTIONS	Uncertain – ready	Focused – confident	Concerned – supported	Relieved – satisfied	Confident – accomplished
PAIN POINTS	Unclear or inconsistent incident-handling process.	Incomplete information or incorrect classification can cause delays and wrong routing.	Knowledge may be difficult to find; manual escalation can lose context; related records may be disconnected.	Resolution documentation may be incomplete.	Useful solutions may go unused; related-record visibility is limited; validation can be inconsistent.
OPPORTUNITIES	Standardize the incident lifecycle and user guidance.	Use consistent required information and classification fields.	Use Knowledge / Agent Assist; standardize escalation; connect changes and child incidents to the parent incident.	Standardize Cause, Resolution, Resolution Details and closure information.	Convert useful resolutions into Knowledge Articles and standardize validation/testing steps.

Journey insight: The journey should minimize incomplete information and manual routing while preserving context across knowledge, Level 2 support, emergency changes, child incidents, resolution, SLA and validation. This directly supports the project's goal of a structured, connected and traceable Incident Management lifecycle.